

FFT Transform

EEG Preprocessing Method — Summary & Validation

1) What FFTTransform does (short & clear)

FFTTransform converts EEG signals from the time domain into a frequency-domain representation by estimating the Power Spectral Density (PSD) using Welch's method. The output shows how signal power is distributed across frequency bins (Hz) for each channel, enabling band-level analysis (e.g., alpha, beta).

Implementation note (MNE usage)

This implementation uses **MNE** (*mne.time_frequency.psd_array_welch*) to compute Welch PSD in a standard, well-tested way. It returns frequency bins in **Hz** (*frequency_hz*) and PSD values per channel (units: V^2/Hz , or $\log_{10}(V^2/Hz)$ if enabled).

2) Why FFTTransform is useful in EEG

Many EEG analyses rely on frequency information (delta/theta/alpha/beta rhythms, line noise detection, and feature extraction). Welch PSD provides a stable spectrum by averaging across shorter overlapping segments, reducing variance compared to a single FFT. This makes frequency peaks easier to interpret and more reliable for downstream tasks.

3) Validation

The next page shows a before/after visualization using the same signal in time domain and its PSD after FFTTransform.

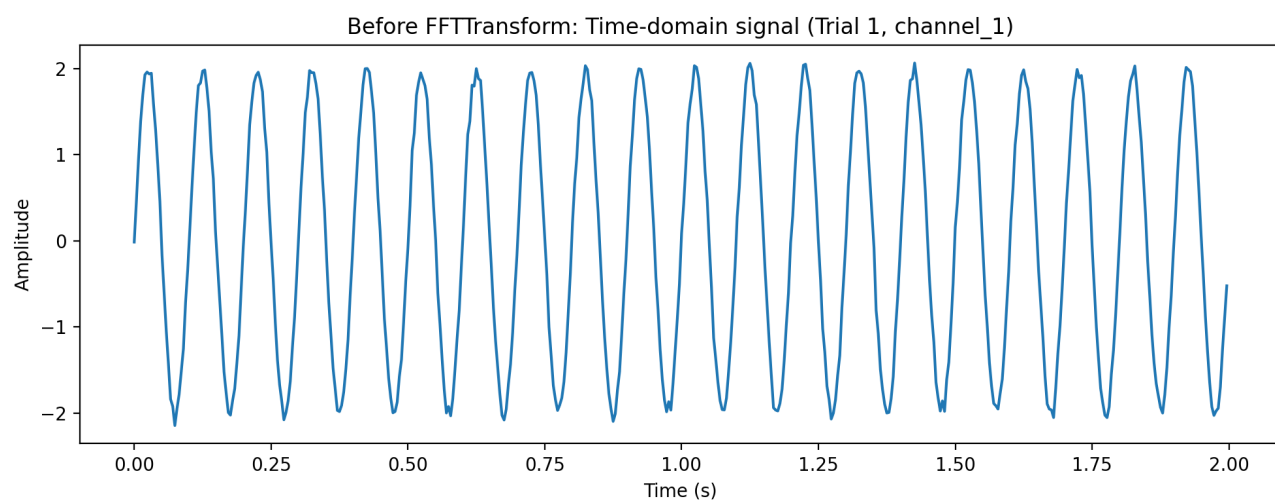


Figure 1 — Time-domain signal before FFTTransform (Trial 1, channel_1).

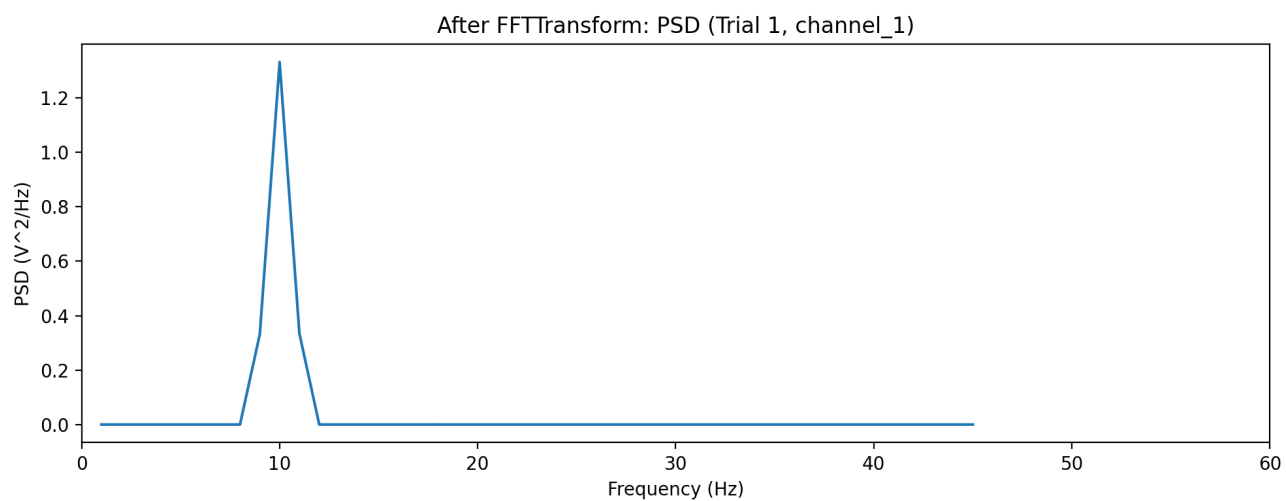


Figure 2 — PSD after FFTTransform (Welch) showing a dominant peak at ~10 Hz (Trial 1, channel_1).